LOCAL QUASI-FINITENESS OF A REPRESENTABLE MORPHISM IS EQUIVALENT TO DISCRETENESS OF ALL FIELD FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The statement that a locally quasi-finite morphism of algebraic spaces has discrete fibres over fields. Invoked inside the counted proof (source lines 4929-4932) as a cross-chapter pointer. Gives discreteness of the fibre over a field from local quasi-finiteness.
- The criterion for checking a property of morphisms of algebraic stacks along a smooth surjective cover. Invoked inside the counted proof (source lines 4936-4940) as a cross-chapter pointer. Reduces local quasi-finiteness of the morphism of stacks to that of a base change covering it.
- The characterization of local quasi-finiteness of a morphism of algebraic spaces by local finite type plus discreteness of field fibres. Invoked at the end of the counted proof (source lines 4945-4948) as a cross-chapter pointer. Concludes local quasi-finiteness of the base-changed morphism.

Source reference. Original reference spaces-morphisms-lemma-locally-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4931–4931. Navigation pointer only: the counted text refers to the Stacks item spaces-morphisms-lemma-locally-quasi-finite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference stacks-properties-lemma-check-property-covering: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 479–479. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-lemma-check-property-covering, which belongs to another chapter of the

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Source reference. Original reference stacks-properties-section-properties-morphisms: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 248–248. Navigation pointer only: the counted text refers to the Stacks item stacks-properties-section-properties-morphisms, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Assume f is representable by algebraic spaces. The following are equivalent*

(1) *f is locally quasi-finite (as in Properties of Stacks, Section stacks-properties-section-properties-morphisms), and*
(2) *f is locally of finite type and for every morphism $\mathrm{Spec}(k) \to \mathcal{Y}$ where k is a field the space $|\mathrm{Spec}(k) \times_{\mathcal{Y}} \mathcal{X}|$ is discrete.*

Proof. Assume (1). In this case the morphism of algebraic spaces $\mathcal{X}_k \to \mathrm{Spec}(k)$ is locally quasi-finite as a base change of f. Hence $|\mathcal{X}_k|$ is discrete by Morphisms of Spaces, Lemma spaces-morphisms-lemma-locally-quasi-finite. Conversely, assume (2). Pick a surjective smooth morphism $V \to \mathcal{Y}$ where V is a scheme. It suffices to show that the morphism of algebraic spaces $V \times_{\mathcal{Y}} \mathcal{X} \to V$ is locally quasi-finite, see Properties of Stacks, Lemma stacks-properties-lemma-check-property-covering. The morphism $V \times_{\mathcal{Y}} \mathcal{X} \to V$ is locally of finite type by assumption. For any morphism $\mathrm{Spec}(k) \to V$ where k is a field

$$\mathrm{Spec}(k) \times_V (V \times_{\mathcal{Y}} \mathcal{X}) = \mathrm{Spec}(k) \times_{\mathcal{Y}} \mathcal{X}$$

has a discrete space of points by assumption. Hence we conclude that $V \times_{\mathcal{Y}} \mathcal{X} \to V$ is locally quasi-finite by Morphisms of Spaces, Lemma spaces-morphisms-lemma-locally-quasi-finite. □